MPSUNet: A Deep Learning-Based Segmentation Framework for Methane Plume Detection With Space-Based Hyperspectral and Multispectral Imagery

Cheng Chen, Meng Fan, Zhibao Wang, *Member, IEEE*, Menglei Liang, Jinhua Tao, and Liangfu Chen

***Abstract*— Methane is a potent greenhouse gas, and its accurate detection is critical for addressing global climate change. Although remote sensing has been a crucial technique for understanding the spatial distribution and temporal dynamics of methane emissions, it is still urgently needed to automate the identification of methane emission plumes and effectively deconvolve the signal from background noise. In this study, we propose the methane plume segmentation UNet (MPSUNet) to achieve precise segmentation of methane plumes from remote sensing imagery. The MPSUNet incorporates the pyramid squeeze attention (PSA) module to enhance feature representation and employs a joint loss function combining Dice Loss and Focal Loss to address challenges such as class imbalance and noisy data. A novel dataset, MPDataset, was constructed using Earth surface mineral dust source investigation (EMIT) methane enhancement and RGB radiance data, providing 4172 high-quality samples for model training and evaluation. Our results show that MPSUNet achieves a mean intersection over union (MIoU) of 78.20%, mean precision of 80.78%, recall of 71.11%, and mean pixel accuracy (MPA) of 85.41% on the complete four-channel MPDataset. Compared with seven classical segmentation models, the most improvements of MPSUNet in MIoU, MPrecision, recall, and MPA reach up to 5.33%, 12.28%, 18.04%, and 8.94%, respectively. Notably, the integration of RGB channels enhances the segmentation of small and intricate plume structures. Cross-dataset evaluation using Sentinel-2 data further validates the model's robustness, achieving an MIoU of 77.65% and an MPA of 83.61%. Generally, the proposed MPSUNet model marks a substantial performance in methane detection, which provides a robust technical framework for global-scale methane emission monitoring as emphasized by global climate agreements.**

***Index Terms*— Methane plume, methane plume segmentation UNet (MPSUNet), pyramid squeeze attention (PSA), remote sensing.**

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I. Introduction

METHANE (CH_4) is primarily emitted from both natural sources (e.g., wetlands and oceans) and anthropogenic activities (e.g., agriculture, oil-and-gas extraction, waste management, and coal mining) [1]. It is a powerful greenhouse gas (GHG) with a global warming potential (GWP) 81 times greater than that of carbon dioxide (CO_2) over a 20-year period timeframe [2], [3], [4]. Despite methane's relatively short atmospheric lifetime of approximately a decade, it exerts a disproportionately large and immediate impact on global warming due to its high radiative forcing. Targeted reduction of methane emissions offers a significant opportunity to slow the rate of climate warming, complementing long-term strategies aimed at addressing CO_2. The urgent need to reduce methane emissions is underscored by the goals outlined in the 2015 Paris Agreement [5]. Nearly 160 countries have now joined the Global Methane Pledge, committing to collectively reduce methane emissions by at least 30% below 2020 levels by 2030. Consequently, the accurate identification and monitoring of these emission sources on a global scale is very important, particularly the precise detection of methane plumes.

Satellite observation offers significant advantages in providing comprehensive global coverage, consistent temporal monitoring, and real-time data collection, which is crucial for understanding the spatial distribution and temporal dynamics of methane emissions. Space-based methane retrieval generally depends on spectrally resolved observations of solar radiation reflected from the Earth's surface in the shortwave infrared (SWIR) spectrum with a range from approximately 1600 to 2500 nm. In recent years, advancements in satellite technology and remote sensing techniques have enhanced precise methane detection with higher resolution and sensitivity. The primary satellite mission for global methane monitoring is

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the TROPOspheric Monitoring Instrument (TROPOMI) [6] onboard the Copernicus Sentinel-5 Precursor (S5P) satellite. Although TROPOMI offers high spectral resolution in the SWIR range that enables the precise determination of methane column concentration data (XCH_4), its relatively coarse spatial resolution (5.5 × 7 km) limits its ability to detect small-scale points emissions. Lauvaux et al. [7] employed TROPOMI to investigate oil-and-gas methane ultraemitters, representing as much as 8%∼12% of global methane emissions from oil-and-gas production methane emissions. Currently, numerous studies have relied on hyperspectral and multispectral satellites with high spatial resolution to capture methane emission plumes even from small emitted sources such as specific industrial facilities, gas wells, or landfills. Such point-source methane-sensing satellites include the German Environmental Mapping and Analysis Program (EnMAP) [8], the Chinese Gaofen 5 (GF5) [9], the Italian PRecursore IperSpettrale della Missione Applicativa (PRISMA) [10], the U.S. Landsat-8/9 [11], [12], the European Sentinel-2 [13], and the commercial GHGSat-C/D [14] and WorldView-3 [15]. This enhanced capability allows for improved tracking of emission sources, which is crucial for targeted mitigation efforts and regulatory enforcement. However, the high spatial resolution of the payload needs to sacrifice the spectral details, especially for multispectral satellites like Sentinel-2 with only a limited number of bands. This tradeoff limits the ability to precisely capture the unique spectral signatures of methane, as fewer bands are available to isolate methane's characteristic absorption features. Furthermore, high spatial resolution increases the impact of surface noise in the detected signal of the satellite. The finer spatial details provided by high-resolution imagery can also result in stronger interference from heterogeneous land surfaces, such as vegetation, water bodies, or urban structures, which introduce additional noise into the methane retrieval process. This effect becomes particularly pronounced when detecting smaller emissions or emissions in less ideal conditions, as the signal-to-noise ratio decreases. Consequently, the accuracy of methane retrieval from such satellite data is compromised, and the presence of anomalous noisy pixels becomes more common, making it challenging to distinguish regions with real high methane concentrations from other areas with anomalous high values.

To track facility-level point-source methane emissions in near-real-time, it is essential to automate the identification of methane emission plumes and effectively deconvolve the signal from background noise. The rapid advancement of deep learning techniques, particularly in computer vision and remote sensing image analysis, has demonstrated their growing potential for accurately detecting and localizing methane plumes in high-resolution satellite-derived methane enhancement images, even in complex and noisy backgrounds. Deep learning models can be trained to recognize the specific spatial and temporal features of methane plumes, allowing for automated detection without requiring manual intervention. For example, Radman et al. [16] generated a benchmark dataset by integrating simulated methane plumes with background noise extracted from real Sentinel-2 images and designed an end-to-end procedure for automatic detection of methane plumes based on a deep learning strategy. Similarly, Jongaramrungruang et al. [17] developed a convolutional neural network (CNN), MethaNet, which can directly predict methane point-source emissions from high-resolution 2-D plume images, without relying on additional local measurements such as background wind speeds. Joyce et al. [18] employed a UNet model for plume mask detection, incorporating an additional 1 × 1 convolutional layer with 64 filters to account for methane plumes associated with anomalies in certain SWIR bands of the PRISMA imagery. Wang et al. [19] treated methane leak size classification as a video classification problem by employing three-dimensional CNNs, VideoGasNet, enabling efficient detection and classification of methane emissions over time. Vaughan et al. [20] introduced CH4Net, a simple and flexible UNet architecture for automatically identifying methane super-emitters from Sentinel-2 data, improving both detection accuracy and computational efficiency. Kumar et al. [21] proposed the hyperspectral Mask-RCNN (H-mrcnn) model to efficiently analyze large-scale hyperspectral data and autonomously detect methane plumes. H-mrcnn combines principled statistics, signal processing, and deep neural networks to address detection challenges associated with small pixel footprint signatures. Sullivan et al. [22] applied fully convolutional networks (FCNs) to imaging spectrometer data of Airborne Visible and InfraRed Imaging Spectrometer Next Generation (AVIRIS-NG) for methane plume detection and delineation. Recent studies have demonstrated how advanced neural architectures tackle similarly complex remote sensing tasks. For instance, Song et al. [23] introduced a visual-modified attention network (VMAN) to optimize the Transformer architecture, effectively filtering noise and enhancing cross-modal reasoning. Zhang et al. [24] presented a dual-branch sparse self-learning framework with instance binding augmentation, achieving state-of-the-art adversarial detection in remote sensing images. Jing et al. [25] proposed the Hypergraph BiFormer, which employs hypergraph convolution to capture high-order relationships for semantic segmentation of high-resolution images. Liu et al. [26] developed a dual-domain network tailored for hyperspectral super-resolution, leveraging both spatial–spectral and frequency-domain features. Meanwhile, Wei et al. [27] introduced a spatiotemporal feature fusion network with guide aggregation to detect changes in multitemporal images. These studies collectively highlight the transformative impact of deep learning in automating methane plume detection, providing a scalable and robust solution for real-time methane monitoring from space-based platforms.

Despite the significant advancements in methane plume detection using deep learning models, several challenges persist. One of the primary obstacles is the acquisition of large-scale, high-quality annotated datasets, which are essential for training robust models. The scarcity of such datasets limits the generalization capability of these models across diverse real-world scenarios. Moreover, the spatial morphology of methane plumes is highly variable, often factors such as topography, atmospheric conditions, and human activities. This variability makes it challenging to develop models that can maintain consistent performance across different environments

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and emission scenarios. Effectively capturing both fine-grained details and the broader spatial context of methane plumes requires the refinement of multiscale information fusion and feature extraction strategies.

Hence, leveraging the capability of deep learning and attention mechanisms to precisely identify and segment methane plume features, this study proposes a methane plume segmentation UNet (MPSUNet) framework. Our main contributions can be summarized as follows.

1) An advanced segmentation model named MPSUNet is introduced by augmenting the classic UNet architecture with the pyramid squeeze attention (PSA) module embedded within skip connections, enhancing the model's capability to capture multiscale spatial features and effectively suppress irrelevant background noise, thereby improving segmentation accuracy and robustness in detecting intricate methane plumes.
2) A composite loss function combining Dice Loss and Focal Loss is proposed and rigorously validated through comprehensive ablation studies, specifically addressing issues of class imbalance and boundary delineation. The integration of this optimized loss function results in enhanced segmentation performance, especially in scenarios characterized by intricate plume structures.
3) Extensive experiments conducted on the newly constructed MPDataset demonstrate the superior performance of MPSUNet relative to several classical and state-of-the-art deep learning segmentation models. The proposed method achieves notable metrics, including MIoU of 78.20%, precision of 80.78%, recall of 71.11%, and MPA of 85.41%, thus validating its effectiveness and reliability. Furthermore, cross-validation conducted on Sentinel-2 datasets confirms the model's robust generalization capabilities, underscoring its practical applicability and adaptability to diverse remote sensing environments.

Section II details the study area and data sources. Section III describes the methodology. Section IV presents experimental details, results, and discussion. Section V is the conclusion.

II. Data

A. EMIT Data

The Earth surface mineral dust source investigation (EMIT) [28] was launched aboard the International Space Station (ISS) in 2022. The imaging spectrometer EMIT instrument measures the spectral range from 380 to 2500 nm with 7.4-nm spectral sampling and has an 11° field of view (80 km) with a 60-m spatial resolution. The EMIT is a pioneering mission by NASA, which aims to use the spectral signatures of minerals observed across the Earth's arid and semi-arid lands containing dust sources to update the soil composition of advanced Earth system models (ESMs) to better understand and reduce uncertainties in mineral dust aerosol radiative forcing at the local, regional, and global scale.

Although the primary objective of the EMIT instrument is to study mineral dust, its hyperspectral capabilities enable it to detect the characteristic absorption features of methane around the 2300-nm absorption band. By distinguishing methane plumes from other atmospheric and surface features, EMIT can identify methane sources at the facility level and track their temporal variations. Here, the study area spans the world's major arid and semi-arid regions between 52°N and 52°S. These regions are not only significant global dust sources but also key areas for methane emissions. This study employs three types of EMIT data products, including L1B At-sensor Calibrated Radiance and Geolocation data (EMITL1BRAD), L2B methane enhancement data (EMITL2BCH4ENH), and L2B Estimated Methane Plume Complexes data (EMITL2BCH4PLM). These datasets, which cover global arid and semi-arid regions from 52°N to 52°S, provide high-precision spectral radiance, methane enhancement information, and methane plume mask, respectively. They can be available on the NASA Earthdata website (https://search.earthdata.nasa.gov/). EMITL1BRAD product provides at-sensor calibrated radiance values along with observation data in a spatially raw, nonorthocorrected format.

EMITL2BCH4ENH product is a total vertical column enhancement estimate of methane in parts per million meter (ppm·m) based on an adaptive matched filter approach. To reduce the risk of false positives, all EMITL2BCH4ENH data undergo a manual review (or identification and confirmation) process before being designated as a plume complex. The EMITL2BCH4PLM product offers verified methane plume complexes through manual auditing of EMITL2BCH4ENH to confirm plume locations and boundaries, ensuring reliable detection results.

B. Dataset Construction

In this study, an EMIT methane plume dataset (MPDataset) was generated to train the MPSUNet model for the automated detection of methane plumes. Based on the methane plumes identified in the EMITL2BCH4PLM product, corresponding images were extracted from both the EMITL2BCH4ENH methane enhancement data and EMITL1BRAD radiance data. Negative samples were selected from the EMITL2BCH4ENH and EMITL1BRAD datasets by identifying dates and locations without methane plumes in the EMITL2BCH4PLM product. We got 1077 positive methane plume samples and 1009 negative ones during the period from August 9, 2022, to April 19, 2024.

Here, with a spatial resolution of 60 m and a size of 512 × 512 pixels, each sample is composed of four data channels. The first three channels correspond to the red, green, and blue (RGB) bands from the EMITL1BRAD radiance data, with central wavelengths of 447.7, 537.0, and 641.3 nm, respectively. The last channel is the corresponding methane enhancement data extracted from the EMITL2BCH4ENH product, which provides critical information on methane concentrations over the observed area. This combination of methane enhancement and visible-spectrum radiance enables the model to capture both the surface and methane plume characteristics.

Moreover, it is noted that the recognizability of methane plumes was reduced due to the large magnitude and wide valid range of the methane enhancement values in the EMITL2BCH4ENH product. In this study, a normalization

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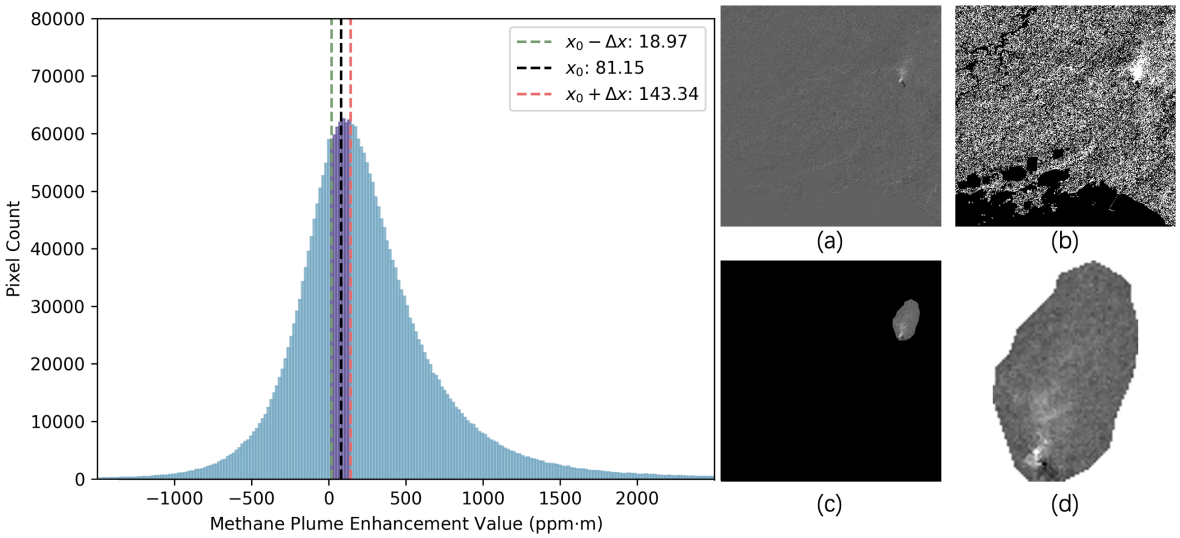

Fig. 1. Histogram illustrating the distribution of methane plume enhancement values across all pixels in the dataset. The horizontal axis represents the methane plume enhancement value (unit: ppm·m), and the vertical axis indicates the total number of pixels corresponding to each enhancement value. (a) Original image. (b) Fourth channel normalized using a pixel-value-distribution-based method. (c) Methane plume mask. (d) EMITL2BCH4PLM data.

method based on the pixel value distribution was applied to the data in the fourth channel of sample images. First, the histogram distribution of methane enhancement values across all pixels was statistically calculated by using the EMITL2BCH4PLM data corresponding to the 1077 positive methane plume samples [Fig. 1(d)]. To establish a reasonable normalization range, the value corresponding to the peak of the histogram (denoted as x_0) was selected as the center, and a step length of Δx was used to define a range where the number of pixels falling within the interval from $x_0 - \Delta x$ to $x_0 + \Delta x$ accounted for 10% of the total pixel number. Subsequently, the methane enhancement data in the fourth channel of all the methane plume samples were linearly stretched by transforming the pixel values from the original range $[x_0 - \Delta x, x_0 + \Delta x]$ to the new range [0, 255]. For our current dataset, x_0 value was determined to be 81.15 ppm·m, and Δx was set to 62.18 ppm·m were determined. By comparing the results in Fig. 1(a) and (b), this normalization process could effectively enhance the visual distinguishability of methane plumes while standardizing the data distribution. It could also reduce the computational burden, enabling faster model convergence and improving the accuracy of methane plume detection.

For the generation of the methane plume mask shown in Fig. 1(c), the pixel values within the methane plume regions, as indicated by the corresponding EMITL2BCH4PLM data, were preserved from the EMITL2BCH4ENH product, while all pixels outside the plume regions were set to zero. This masked representation retains the original enhancement values only within the plume area, enabling a clearer visualization of methane distribution in the target region. These methane plume masks of positive and negative samples were then labeled as the reference ground-truth image for machine learning training.

Given the limited number of samples derived from the EMIT data, data augmentation is necessary to improve the model's generalization capability. This involves generating additional samples by applying various transformations or perturbations to the existing data, thereby expanding and diversifying the dataset. In this study, we rotated the original sampled data by 180 degrees counterclockwise, constructing a final MPDataset composed of 2154 positive samples and 2018 negative ones. These total 4172 samples of MPDataset were divided into three subsets: a training set containing 3004 images, a validation set with 334 images, and a test set comprising 834 images. A representative sample from MPDataset is shown in Fig. 2.

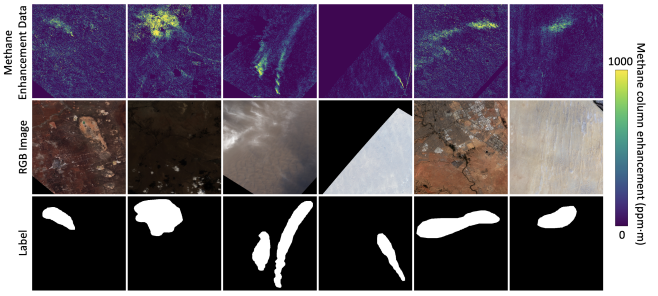

Fig. 2. Representative samples from the MPDataset. From top to bottom: methane enhancement data from EMITL2BCH4ENH, RGB images from EMITL1BRAD, and corresponding plume masks based on EMITL2BCH4PLM.

III. Methodologies

A. MPSUNet Models

A classical UNet [29] architecture consists of two primary components: an encoder and a decoder. The encoder serves as a feature extraction network for progressively downsampling the input to capture essential features at various levels of abstraction. Conversely, the decoder is responsible for mapping the low-resolution features back to the high-resolution image space, facilitating the recovery of fine-grained details. Based on the UNet architecture, we proposed a novel methane plume segmentation network, MPSUNet, as illustrated in Fig. 3. An important modification in our MPSUNet model is

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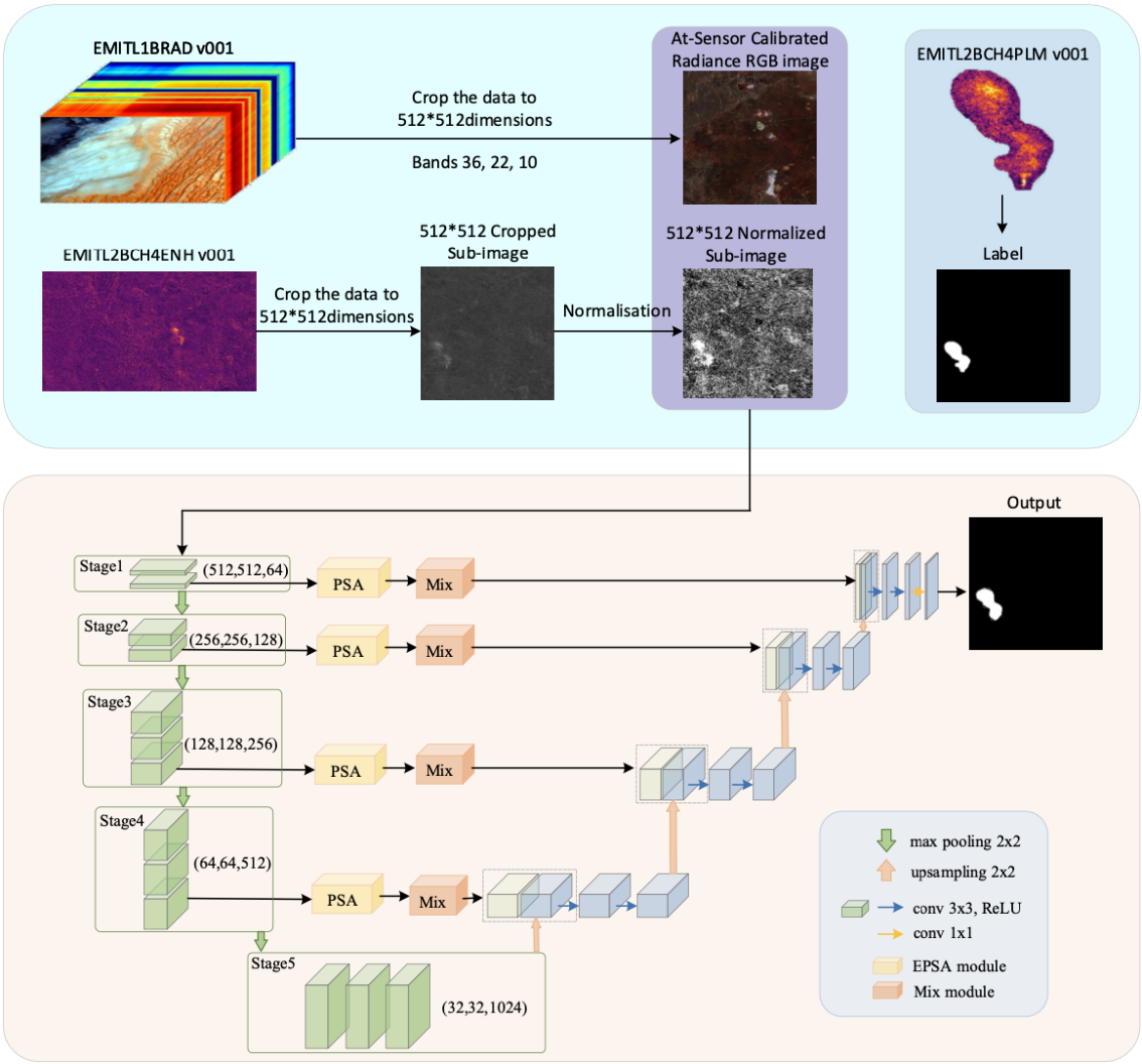

Fig. 3. Diagram of the proposed MPSUNet network architecture. We replace the feature extraction network counterpart of UNet with VGG16, which is divided into five residual modules. The PSA attention mechanism is added in the jump connection branch, after which the feature map is spliced. The decoder maintains the original structure of UNet.

the replacement of the traditional encoder backbone with the Visual Geometry Group with 16-layer deep model architecture (VGG16) [30]. VGG16 is a deep convolutional network renowned for its ability to capture complex and hierarchical features, which is particularly advantageous when target boundaries are blurred or when precise identification of small and challenging objects (e.g., methane plumes) is required. Here, we utilized the first 13 layers of VGG16 to extract feature information at multiple levels. Specifically, the first 13 layers in the encoder part of VGG16 were divided into five stages, and the image size was reduced to 1/16th of the original dimensions. The first two stages consist of two 3×3 convolutional layers, while the remaining three stages are composed of three 3×3 convolutions each. Additionally, to ensure effective feature compression, there are four downsampling operations with pooling window sizes of 2×2 and strides of 2.

To enhance the representation of multiscale features, the PSA module was integrated into the skip connection branches between the encoder and the decoder. This module refines the intermediate features extracted by VGG16, adjusting their weights to focus on the most relevant information for methane plume segmentation while suppressing less useful data. By focusing on these critical features, the PSA module helps to improve the decoder's ability to recover and reconstruct fine details from the low-resolution encoded information. In the decoder part, our MPSUNet model inherits the classical UNet architecture, with four upsampling operations to gradually restore low-resolution features back to their original high-resolution. These upsampled features are then fused with the high-resolution features passed through the skip connections, thereby ensuring that the network retains essential spatial information. Additionally, to better facilitate the flow of information between shallow and deep layers, we adopted an adaptive hybrid operation for feature fusion. This operation replaces the traditional skip connection, enabling more effective integration of features from both the downsampling and upsampling stages.

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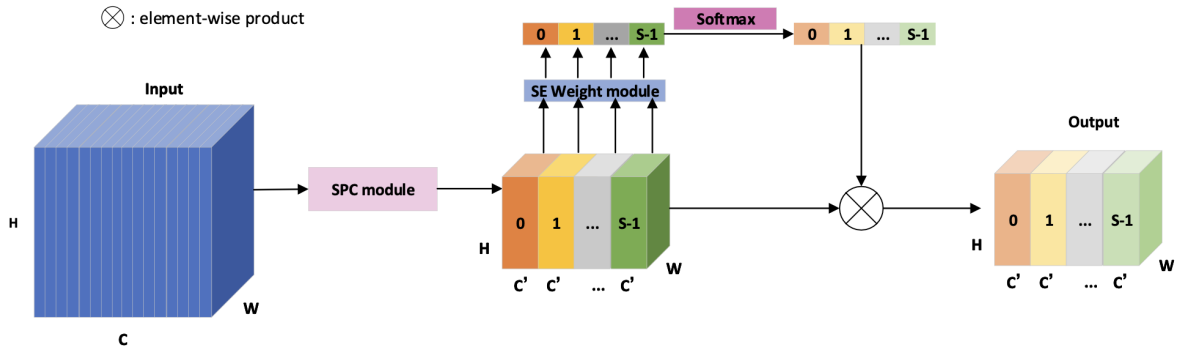

Fig. 4. Structure of the proposed PSA module.

B. PSA Module

PSA leverages a pyramid structure and multiscale convolutions to extract features and perform cross-dimensional interactions, thereby enhancing the network's spatial and channel attention capabilities, as shown in Fig. 4. The PSA module can be divided into several key stages as follows.

Stage 1: Squeeze pyramid concat (SPC) module.

In this stage, the input feature map is processed in parallel by convolution kernels of different sizes to capture spatial information at multiple scales. For the ith scale, the convolutional feature map F_i is computed as

$$F_i = \text{Conv}(k_i \times k_i, G_i)(X), \quad i = 0, 1, \ldots, S-1 \tag{1}$$

where F_i is the feature map at the ith scale, k_i is the size of the ith convolution kernel, G_i denotes the group size for group convolution, X is the input feature map, and S is the total number of convolution kernels. The feature maps from all scales are concatenated along the channel dimension to form a multiscale feature map F

$$F = \text{Cat}\left(\left[F_0, F_1, \ldots, F_{S-1}\right]\right) \tag{2}$$

where "Cat" denotes concatenation along the channel dimension.

Stage 2: Squeeze-and-excitation (SE) block.

To further refine the importance of each channel, global average pooling (GAP) is applied to each feature map to generate global information g_c for the cth channel [31]

$$g_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j) \tag{3}$$

where g_c is the global average pooling result value of for the cth channel, H and W represent the height and width of the feature map, respectively, and $x_c(i, j)$ is the pixel value at position (i, j) in the cth channel. Then, a two-layer fully connected (FC) network is used to generate attention weights w_c for each channel

$$w_c = \sigma(W_1 \delta(W_0(g_c))) \tag{4}$$

where $W_0 \in \mathbb{R}^{(C/r)\times C}$ and $W_1 \in \mathbb{R}^{C\times(C/r)}$ are the weight matrices of the FC layers, δ is the ReLU activation function, σ is the sigmoid activation function, and w_c is the resulting channel attention weight. Then, the SE module is used to extract the channel attention weights from the multiscale feature maps F_i obtained above, which is formulated as

$$Z_i = \text{SE}(F_i), \quad i = 0, 1, 2, \ldots, S-1 \tag{5}$$

where Z_i represents the attention weights corresponding to F_i. To further achieve efficient interaction of attention information, the cross-dimension vectors are fused, and a concatenation operation is applied to obtain the complete multiscale attention weight vector Z

$$Z = Z_0 \oplus Z_1 \oplus \cdots \oplus Z_{S-1}. \tag{6}$$

Stage 3: Re-calibrated weight.

Soft attention adaptively selects spatial scales across channels, guided by Z_i. The soft attention weights are computed as

$$\text{att}_i = \frac{\exp(Z_i)}{\sum_{j=0}^{S-1} \exp\left(Z_j\right)} \tag{7}$$

where att_i is the recalibrated attention weight for the ith scale. To enable interaction between local and global channel attention, the soft weights att_i are re-calibrated using the Softmax operation. The re-calibrated attention weights are then integrated through a concatenation operation to form the complete channel attention vector att

$$\text{att} = \text{att}_0 \oplus \text{att}_1 \oplus \cdots \oplus \text{att}_{S-1}. \tag{8}$$

Stage 4: Element-wise multiplication.

In this step, the att_i is applied to the corresponding F_i via element-wise multiplication, resulting in an adjusted feature map Y_i

$$Y_i = F_i \odot \text{att}_i \tag{9}$$

where $\odot$ denotes element-wise multiplication and Y_i represents the adjusted feature map for the ith scale.

Stage 5: Feature concatenation for final output.

The weighted feature maps Y_i at all scales are concatenated to generate the final output of the PSA module

$$\text{Out} = \text{Cat}\left(\left[Y_0, Y_1, \ldots, Y_{S-1}\right]\right). \tag{10}$$

This concatenated output is then passed on to the subsequent layers of the network for further processing. By effectively integrating multiscale spatial information and channel-wise attention, the PSA module enhances the model's feature

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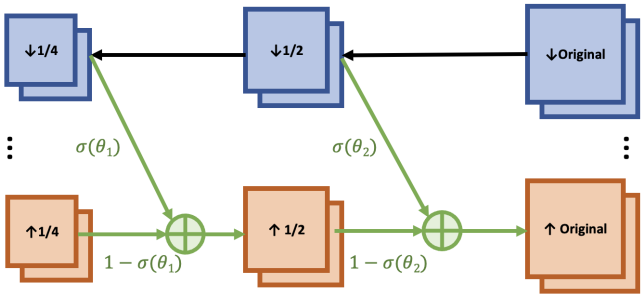

Fig. 5. Adaptive mixup. The first and second rows are downsampling and upsampling operations, respectively.

representation while maintaining relatively low computational complexity. This allows the model to focus more effectively on key features, leading to improved performance in tasks such as methane plume detection, where spatial and contextual information are critical.

C. Adaptive Mixup

As the depth of a neural network increases, the ability of shallow layers to retain fine-grained information, such as edges and contours, gradually degrades. Traditional approaches (e.g., skip connections) have been employed to fuse shallow and deep layers to combine low-level details with high-level semantic information. However, in traditional skip connections, the contribution of each layer's features is fixed and remains the same throughout the network. This static proportion often fails to adapt to the degradation of shallow features due to different inputs.

To address this limitation, this study introduced an adaptive feature fusion mechanism, termed adaptive mixup, which utilizes a learnable factor $\sigma(\theta)$ to adaptively fuse features from downsampling and upsampling layers, as shown in Fig. 5. The learnable factor is obtained by applying a sigmoid function to the network's parameters θ, allowing the model to automatically adjust the contribution of each feature map based on its relative importance. This adaptive fusion mechanism facilitates the preservation of crucial shallow information (e.g., edges and texture), while effectively integrating higher-level semantic features from deeper layers.

Mathematically, the output of the adaptive hybrid operation is given by

$$F_{\text{mix}} = \text{Mix}(f_s, f_d) = \sigma(\theta) * f_s + (1 - \sigma(\theta)) * f_d \tag{11}$$

where f_s and f_d represent the features from the shallow and deep layers, respectively. $\sigma(\theta)$ is a learnable factor determined by applying the sigmoid function on parameter θ.

D. Loss Function

To address both class imbalance and region-matching challenges in segmentation tasks, a combination of Dice Loss and Focal Loss was employed as the joint loss function. This hybrid approach ensures that the model is capable of handling complex scenarios by balancing the precision of foreground–background segmentation with the difficulty of classifying minority classes.

Dice Loss [32] is particularly effective in enhancing segmentation accuracy by maximizing the overlap between ground-truth labels and the predicted segmentation masks. It can be formulated as

$$L_{\text{Dice}} = 1 - \frac{2\sum_i y_i \hat{y}_i}{\sum_i y_i + \sum_i \hat{y}_i} \tag{12}$$

where y_i is the ground-truth label and $\hat{y}_i$ is the predicted probability. The Dice coefficient emphasizes maximizing the intersection between predicted and true values while minimizing false positives and false negatives.

Focal Loss [33], on the other hand, is designed to address the class imbalance by increasing the loss weight for hard-to-classify samples. This reduces the effect of easy negative samples that dominate the training process, enabling the model to focus on challenging, often minority, class examples. The Focal Loss is expressed as

$$L_{\text{Focal}} = -\sum_i \alpha \left(1 - \hat{y}_i\right)^{\gamma} y_i \log\left(\hat{y}_i\right) \tag{13}$$

where α is the balancing factor used to adjust the weight between the classes and γ is a parameter that controls the rate at which easy examples are down-weighted. Higher values of γ increase the emphasis on hard-to-classify instances.

The final joint loss function is a weighted sum of Dice Loss and Focal Loss, formulated as

$$L = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{Focal}} \tag{14}$$

where λ_1 and λ_2 are tunable parameters for balancing the influence of Dice Loss and Focal Loss. In our experiments, we set $\lambda_1 = \lambda_2 = 1$, resulting in an equal contribution from both losses to optimize the model performance.

E. Evaluation Indicators

To evaluate the performance of the proposed MPSUNet method, four widely used segmentation metrics were employed: mean intersection over union (MIoU), mean precision (MPrecision), recall, and mean pixel accuracy (MPA), as shown in (15)–(18). MIoU quantifies the accuracy of the segmentation by easing the overlap between predicted and true regions. MPrecision evaluates the model's ability to correctly identify methane plume pixels by calculating the precision at each class and averaging across all classes. Recall measures the model's ability to correctly identify all methane plume pixels by computing the ratio of true positives to the total number of ground truth positive pixels (true positives + false negatives). MPA provides an overall accuracy measure by computing the mean accuracy per class, which is defined as the ratio of true positives to the sum of true positives and false negatives

$$\text{MIoU} = \frac{1}{N}\sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{15}$$

$$\text{MPrecision} = \frac{1}{N}\sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \tag{17}$$

$$\text{MPA} = \frac{1}{N}\sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \tag{18}$$

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TABLE I
SEGMENTATION RESULTS OF DIFFERENT METHODS ON THE METHANE ENHANCEMENT DATASET. BOLD INDICATES THE BEST RESULTS. IN THE METHOD COLUMN, "+" BEFORE INDICATES AN ENCODER, "+" AFTER INDICATES A DECODER

Model	MIoU(%)	MPrecision(%)	Recall(%)	MPA(%)
VGG16+UNet [29]	75.34	75.25	61.83	77.24
MobileNetV3+PspNet [35]	73.26	74.49	53.48	76.67
MobileNetV3+DeeplabV3plus [36]	76.95	76.68	67.10	82.56
HrNet [37]	72.80	73.26	52.89	76.38
Segformer [38]	71.93	68.26	51.54	75.69
SDSCUNet [39]	77.11	78.54	68.47	83.73
Samba [40]	76.41	76.51	65.11	80.95
MPSUNet	**78.06**	**79.84**	**70.89**	**85.33**

where TP_i are the true positives, indicating the number of pixels correctly predicted as category i. FP_i are the false positives, indicating the number of pixels incorrectly predicted as category i when they do not actually belong to category i. TN_i are the true negatives, representing the number of pixels correctly predicted as other classes. FN_i are the false negatives, indicating the number of pixels that actually belong to category i but are predicted as other categories.

IV. RESULTS AND DISCUSSION

Our experiments were conducted by using two distinct datasets for model training. One is a single-channel dataset with only methane enhancement data from the first channel of the MPDataset, focusing solely on the methane increase information. The other one is the complete MPDataset with four-channel data, providing a broader range of features for model training. The experimental setup employed in this study utilized the Ubuntu operating system. Our MPSUNet model was implemented using Python 3.10 and PyTorch 1.13.0, with training carried out on a GeForce RTX 2080Ti GPU. The batch size was set to 4, and the initial learning rate was configured to 0.0001. The model was trained over 500 epochs by using the Adam optimizer with exponential decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.999$. To prevent gradient explosion, a cosine annealing strategy was employed to adjust the learning rate dynamically throughout the training process. To further evaluate the performance of our MPSUNet model, several state-of-the-art models, including VGG16 + UNet, MobileNetV3 + PspNet [34], [35], MobileNetV3 + DeeplabV3plus [36], HrNet [37], Segformer [38], SDSCUNet [39], and Samba [40]. This comparison was essential to benchmark the efficacy of the MPSUNet model in the context of methane plume detection and monitoring.

A. Methane Plume Detection Based on the Single-Channel Methane Enhancement Dataset

Comparative experiments were first conducted on the single-channel methane enhancement dataset. The quantitative segmentation performance results of various models are detailed in Table I. Specifically, MPSUNet achieved the highest MIoU (78.06%), MPrecision (79.84%), recall (70.89%), and MPA (85.33%), indicating superior overall performance in both detecting methane plumes and minimizing errors. In comparison, SDSCUNet also performed strongly, with the second-highest MIoU (77.11%) and MPA (83.73%), reflecting its strong segmentation performance. However, its MPrecision (78.54%) and recall (68.47%) were slightly lower than those of MPSUNet, suggesting it was less effective in minimizing false positives and detecting all relevant plume pixels. MobileNetV3 + DeeplabV3plus and Samba also demonstrated competitive performance, with MobileNetV3 + DeeplabV3plus achieving an MIoU of 76.95% and MPrecision of 76.68%, indicating relatively high accuracy in methane plume detection. However, these models fell short in terms of recall and MPA when compared to MPSUNet and SDSCUNet. Similarly, Samba showed relatively high MPrecision (76.51%) but lower recall (65.11%), indicating it struggled more with detecting all methane plume pixels. VGG16 + UNet, while a commonly used architecture, yielded an MIoU of 75.34%, demonstrating decent performance, but its lower recall (61.83%) suggests that it missed a larger proportion of methane plume pixels. MobileNetV3 + PspNet and HrNet performed the least favorably, with MIoU values of 73.26% and 72.80%, respectively. These models exhibited notably lower recall (53.48% and 52.89%) and MPA (76.67% and 76.38%), indicating that they were less effective in both detecting and accurately classifying methane plumes. The Segformer model, with the lowest performance in all four metrics, achieved an MIoU of 71.93%, MPrecision of 68.26%, recall of 51.54%, and MPA of 75.69%. These results suggest that Segformer struggled the most in accurately detecting and classifying methane plume pixels, performing poorly in both MPrecision and recall. Generally, our MPSUNet achieved satisfactory results in the methane plume recognition task.

Fig. 6 depicts the visualization results of five specific methane plume segmentation cases based on eight models listed in Table I. These results were analyzed to evaluate the performance of each model in detecting methane plumes, especially focusing on the preservation of fine-grained details at the plume boundaries. MPSUNet stands out as the best-performing model across all categories. It provides a detailed and accurate segmentation, with the boundaries of the methane plumes preserved more effectively than with other models. Even the smaller and more challenging plumes are detected with high precision, making this model the most reliable for methane plume detection. For Case 1, models such as VGG16 + UNet, MobileNetV3 + PspNet, MobileNetV3 + DeeplabV3plus, HrNet, Segformer, SDSCUNet, and Samba all exhibit varying degrees of detail loss, with significant portions of the plume boundaries being blurred or

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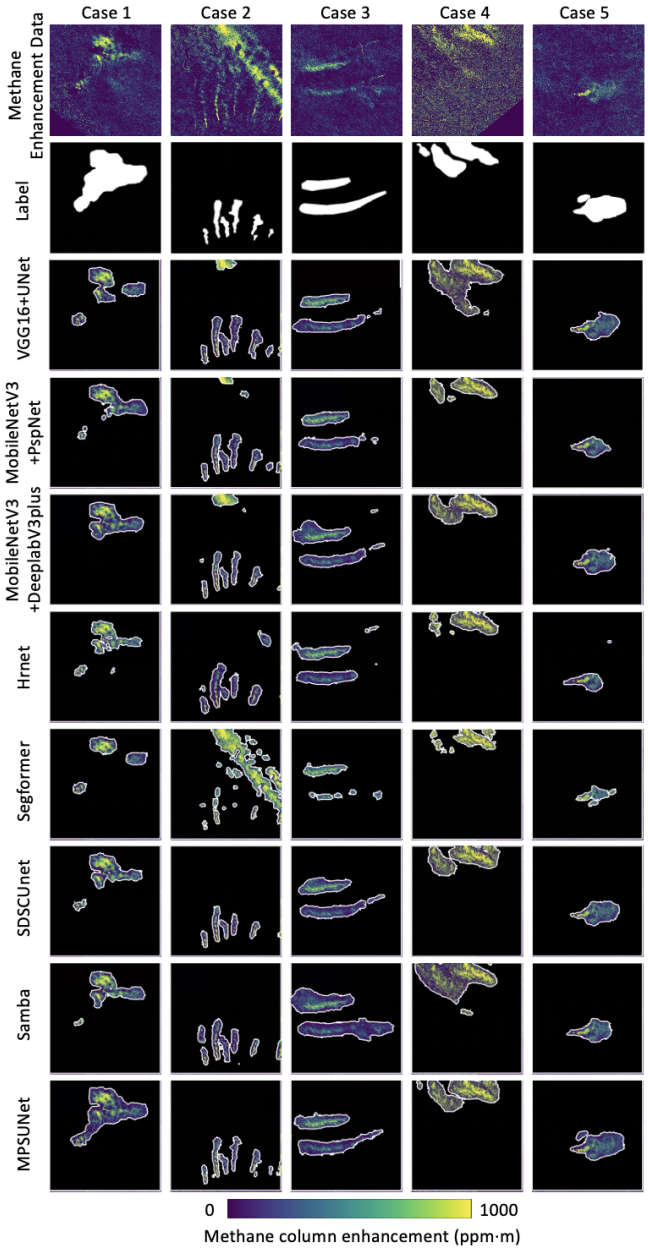

Fig. 6. Methane plume detection results on the single-channel methane enhancement dataset using various methods.

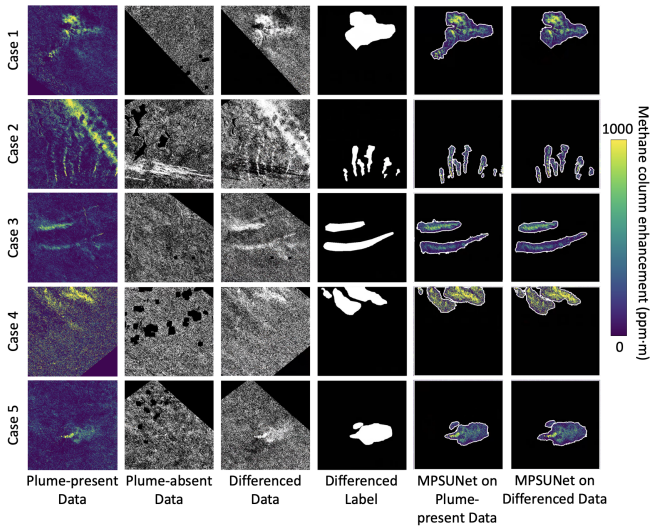

Fig. 7. Prediction results on differenced methane enhancement data.

inadequately captured. For Case 2, false detections are observed in models like VGG16 + UNet, MobileNetV3 + PspNet, MobileNetV3 + DeeplabV3plus, HrNet, and Segformer, while MobileNetV3 + DeeplabV3plus, HrNet, Segformer, SDSCUNet, and Samba fail to identify some small methane plume targets. This highlights the limitations of these models in accurately identifying all plume features, particularly the finer details of smaller plumes. For Case 3, VGG16 + UNet, MobileNetV3 + PspNet, MobileNetV3 + DeeplabV3plus, HrNet, Segformer, and SDSCUNet suffer significant detail loss. Additionally, MobileNetV3 + DeeplabV3plus and HrNet yield false detections, and Samba produces relatively coarse segmentation results. This further underscores the deficiencies of these methods in terms of both MPrecision and recall. For Case 4, the segmentation resulted by VGG16 + UNet's segmentation is notably coarse, and MobileNetV3 + PspNet, MobileNetV3 + DeeplabV3plus, HrNet, Segformer, and SDSCUNet fail to recognize small methane plumes, while Samba suffers from severe detail loss. For Case 5, VGG16 + UNet, MobileNetV3 + PspNet, MobileNetV3 + DeeplabV3plus, HrNet, SDSCUNet, and Samba cannot detect small methane plumes, and Segformer shows substantial detail loss.

Furthermore, to enhance the visibility of plume features by minimizing the background interference, a differencing operation was applied to single-channel methane enhancement data collected from the same region on both plume-present and plume-absent dates. Initially, single-channel methane enhancement data were extracted separately from both the plume-present and plume-absent dates, and a pixel-by-pixel subtraction was performed to generate differenced data. Then, the above MPSUNet model that was trained on the single-channel methane enhancement data was used to predict results on both the original plume-present data and the differenced data. As shown in Fig. 7, although the predictions on plume-present data successfully identified most plume regions, some areas were subject to false detections due to noise and data complexity. In contrast, predictions on the difference-processed data are notably more accurate. Not only did the model more precisely delineate plume regions, but it also significantly reduced both the number of false and missed detections. Therefore, the differencing method further strengthened the methane plume features by mitigating background noise, enabling the MPSUNet model to more accurately identify plume regions during prediction and confirming the effectiveness of this approach.

B. Methane Plume Detection Based on MPDataset

1) Analysis of Model Results: Similar experiment in Section IV-A but for the complete MPDataset was conducted, and corresponding evaluation indicators and visual results are listed in Table II and Fig. 8, respectively. Compared to the methane plume recognition model trained using only the methane enhancement data, the model trained with the

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TABLE II
SEGMENTATION RESULTS OF DIFFERENT METHODS ON THE METHANE PLUME DATASET MPDATASET. BOLD INDICATES THE BEST RESULTS. IN THE METHOD COLUMN, "+" BEFORE INDICATES AN ENCODER, "+" AFTER INDICATES A DECODER

Model	MIoU(%)	MPrecision(%)	Recall(%)	MPA(%)
VGG16+UNet	76.23	77.49	63.27	80.93
MobileNetV3+PspNet	75.64	76.26	62.02	79.55
MobileNetV3+DeeplabV3plus	77.08	79.42	67.87	82.88
HrNet	73.85	73.71	61.83	79.23
Segformer	72.87	68.50	53.07	76.47
SDSCUNet	77.56	80.24	68.87	84.11
Samba	76.80	78.46	65.72	81.78
MPSUNet	**78.20**	**80.78**	**71.11**	**85.41**

complete MPDataset demonstrated an improvement in its ability to identify methane plumes. However, the enhancement was not substantial. Specifically, the MIoU increased from 78.06 to 78.20, MPrecision from 79.84 to 80.78, recall from 70.89 to 71.11, and MPA from 85.33 to 85.41.

These results in Table II also indicate that the MPSUNet model outperforms the other methods in methane plume recognition tasks, offering significant improvements across all key metrics. Compared to the VGG16 + UNet model, MPSUNet demonstrated significant improvements, achieving a 1.97% increase in MIoU, a 3.29% increase in MPrecision, a 7.84% improvement in recall, and a 4.48% increase in MPA. When compared with MobileNetV3 + PspNet, MPSUNet showed even more notable improvements, with increases of 2.56% in MIoU, 4.52% in MPrecision, 9.09% in recall, and 5.86% in MPA. Against MobileNetV3 + DeeplabV3plus, MPSUNet achieved more modest gains, with MIoU improving by 1.12%, MPrecision by 1.36%, recall by 3.24%, and MPA by 2.53%. These results demonstrate that while MPSUNet provides incremental improvements over MobileNetV3 + DeeplabV3plus, the differences are less pronounced compared to some of the other methods. When compared to HrNet, MPSUNet showed notable advances, with increases of 4.35% in MIoU, 7.07% in MPrecision, 9.28% in recall, and 6.18% in MPA. Furthermore, MPSUNet outperformed Segformer by a significant margin, improving MIoU by 5.33%, MPrecision by 12.28%, Recall by 18.04%, and MPA by 8.94%. In comparison with SDSCUNet, MPSUNet achieved modest improvements, with MIoU increasing by 0.64%, MPrecision by 0.54%, recall by 2.24%, and MPA by 1.3%, demonstrating its effectiveness even when compared to a strong baseline model. Finally, compared to Samba, MPSUNet showed improvements of 1.4% in MIoU, 2.32% in MPrecision, 5.39% in recall, and 3.63% in MPA, further establishing its superiority in methane plume detection tasks. As for the visual results of Cases 1–5 in Fig. 9, MPSUNet also consistently provides the highest-quality segmentation results among all 8 models, effectively detecting and delineating methane plumes with greater precision and fewer inaccuracies.

Fig. 9 further presents a comparison of predictions made by MPSUNet on the MPDataset and the single-channel methane enhancement dataset. For Case 1, it is apparent that the use of four-channel data results in predictions that more closely align with the ground-truth labels, particularly in terms of capturing the plume shape and preserving edge details. Cases 2 and 3 indicate that using only single-channel data leads to omissions or misclassifications in certain areas, whereas the four-channel predictions are nearly identical to the labels, with substantially improved boundary delineation. For Cases 4 and 5, the addition of RGB data markedly enhances the model's ability to detect small plume structures, yielding results that closely mirror the annotated masks and effectively distinguishing plume regions from the background. In summary, incorporating RGB data enables more accurate capture of plume morphology and boundary information, thus improving predictive performance in complex scenarios where the plumes may exhibit intricate shapes and subtle features. This demonstrates that the spatial cues provided by RGB channels further strengthen the model's capability to recognize and segment methane plumes with greater precision.

2) Validation Analysis Based on Emission Source and Wind Data: Although the MPSUNet model trained on the MPDataset has demonstrated a strong performance in detecting EMIT methane plumes, some detected plumes are not involved in the EMITL2BCH4PLM product. To further validate the reliability of these detected plumes, additional verification steps were implemented based on emission sources and wind field data. Here, high-resolution Google Earth imagery was employed to assess whether potential emission facilities are located near the detected plumes, allowing for the identification of possible methane emission sources that generate this plume. Wind field data from the Fifth Generation of the European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis (ERA5) product was utilized to determine if the direction of plume dispersion aligned with the prevailing wind direction. This analysis provided further evidence of the plumes' authenticity by confirming that their movement was consistent with the wind pattern. If both conditions (i.e., the proximity of potential emission sources and the alignment of plume direction with the wind) are satisfied, the identified plume was considered unlikely to be a false positive.

Two representative cases detected by the MPSUNet model but not present in the EMITL2BCH4PLM product are shown in Fig. 10. Further analysis revealed that such detected plumes were primarily concentrated around oil tank areas and oil well sites which are emission facilities known to be significant methane emission sources. For Case 1 shown in Fig. 10, the oil tank is situated at the southwestern corner of the detected methane plume. Given the influence of the southwest wind,

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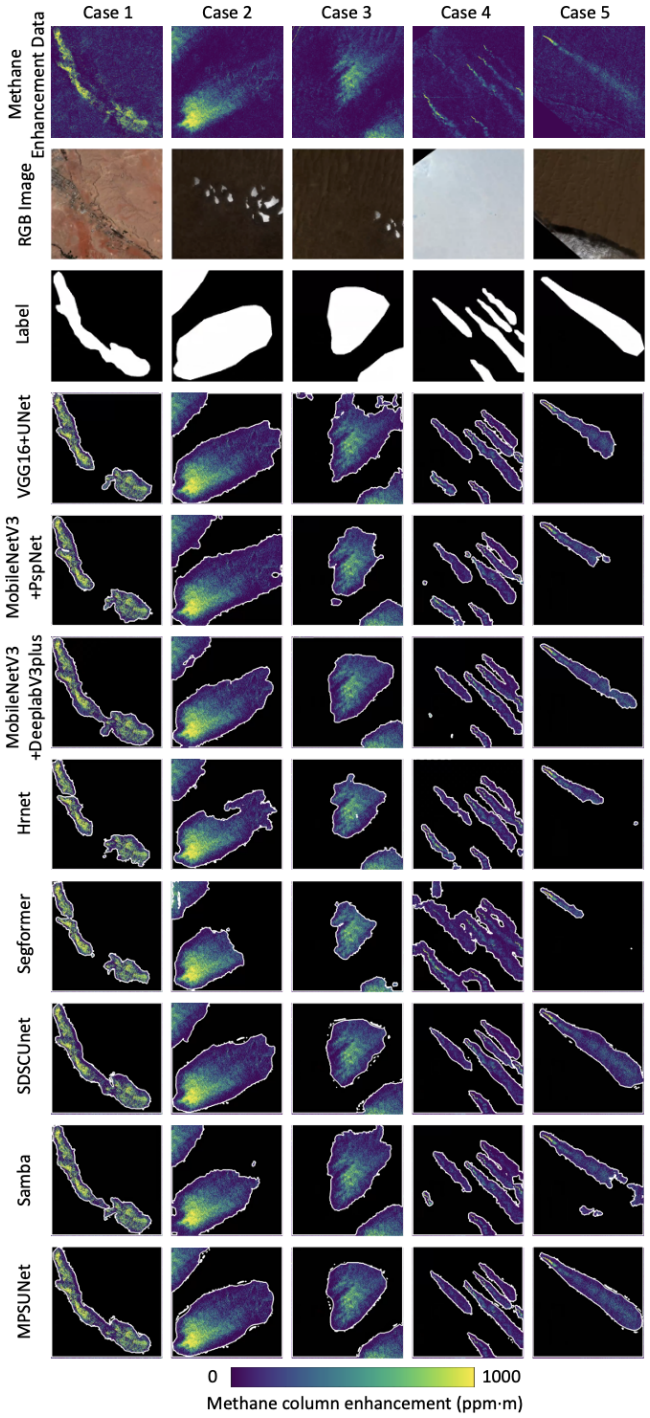

Fig. 8. Methane plume detection results on MPDataset using various methods.

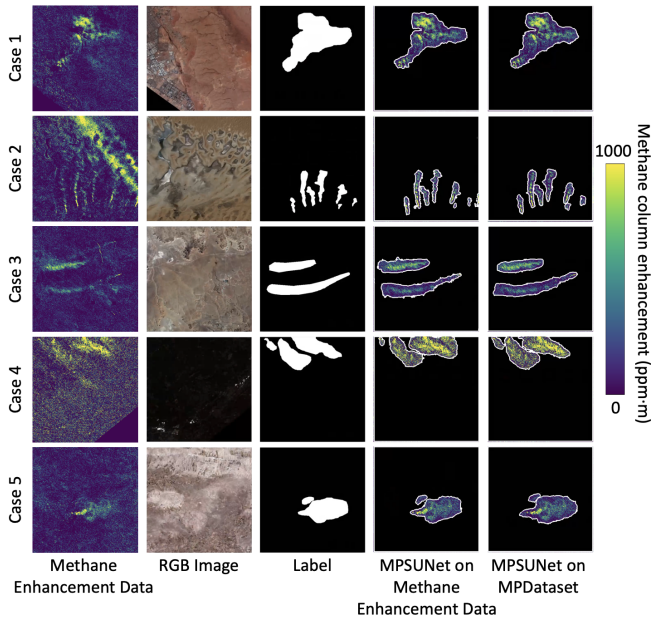

Fig. 9. Comparison of MPSUNet predictions on the single-channel methane enhancement dataset and the MPDataset.

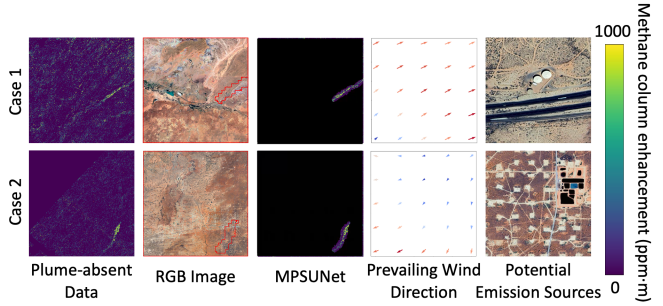

Fig. 10. MPSUNet detection results on EMIT satellite data.

it is expected that the plume would disperse in a northeast direction, which aligns with the plume dispersion detected by the MPSUNet model. Similar to Case 2 shown in Fig. 10, the presence of a significant number of oil wells at the northeastern corner of the detected methane plume, combined with the prevailing northeast wind, further provides strong evidence supporting the validity of the methane plume detection results. These findings suggest that our MPSUNet model still can efficiently detect the distribution of methane plumes even in unlabeled regions. It highlights the practical utility of the MPSUNet model in enhancing the ability to automatically identify methane emissions.

Therefore, when processing a huge number of satellite methane enhancement data, manual identification, and confirmation may fail to smaller methane emission plumes due to the limitations of manual review. In contrast, machine learning-based methane plume recognition techniques can effectively address this issue by systematically analyzing the data, enabling the detection of even small or subtle plumes that might be missed during manual review like for EMIT data.

3) Analysis of Model Efficiency: To evaluate the computational efficiency of each model, floating-point operations (FLOPs), parameter count (Params), inference time, memory usage, and frames per second (FPS) were employed as key metrics. FLOPs reflect the computational complexity of inference, measured in units of 10^9 (denoted as, G), while Params quantify the model's storage requirements in units of 10^6 (denoted as, M). Inference time represents the time required to process a single-input sample, measured in milliseconds (ms). Memory usage indicates the peak GPU memory consumption during inference, measured in megabytes (MB). FPS measures the inference throughput, indicating the number of images processed per second by the model. To ensure a fair comparison

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TABLE III
EFFICIENCY OF VARIOUS MODELS. BOLD INDICATES THE BEST RESULTS. IN THE METHOD COLUMN, "+" BEFORE INDICATES AN ENCODER, "+" AFTER INDICATES A DECODER

Method	FLOPs(G)	Params(M)	Inference Time(ms)	Memory(MB)	FPS
VGG16+UNet	225.99	24.89	32.02	198.29	31.23
MobileNetV3+PspNet	**3.07**	**2.41**	**2.12**	**26.68**	**472.70**
MobileNetV3+DeeplabV3plus	26.45	5.81	4.44	51.03	225.23
HrNet	18.70	9.64	3.82	81.06	261.78
Segformer	6.79	3.72	4.04	34.41	247.52
SDSCUNet	29.95	21.32	10.56	62.58	94.70
Samba	231.37	46.00	49.13	297.15	20.35
MPSUNet	226.05	24.89	46.61	198.62	21.45

across models, all evaluations were conducted using an input image size of $4 \times 512 \times 512$. Table III summarizes the computational efficiency metrics for all eight tested models. Among these, the MobileNetV3 + PspNet model exhibits the lowest overall computational complexity, making it the most efficient model in terms of both FLOPs and Params and achieves a significantly higher FPS (472.7). However, despite its efficiency advantages, the MobileNetV3 + PspNet model typically yields lower segmentation accuracy. Taking VGG16 + UNet as the baseline, it is evident that MPSUNet achieves superior methane plume segmentation performance without substantially increasing the model's computational complexity. Consequently, MPSUNet effectively balances improved segmentation accuracy with computational efficiency comparable to that of the established baseline. Furthermore, compared with the latest Samba model, MPSUNet exhibits lower FLOPs, Params, inference time, and memory usage. Although the complexity of MPSUNet slightly surpasses some lightweight alternative models, it maintains an optimal balance between model complexity and segmentation accuracy. This desirable tradeoff ensures robust and precise methane plume detection without incurring prohibitively high computational demands, thereby affirming its suitability for practical application in large-scale, satellite-based remote sensing scenarios.

4) Analysis of the Attention Module: In semantic segmentation tasks, the integration of attention mechanisms has been shown to substantially enhance feature representation and improve segmentation accuracy. These attention mechanisms can be incorporated into various parts of the network, including the encoder, the decoder, and skip connections. However, the placement of these mechanisms in the network can influence the overall performance in different ways.

The encoder primarily captures global features, but it may fail to capture finer spatial details. Incorporating excessive attention modules at this stage may also increase computational overhead. The decoder mainly focuses on upsampling and feature reconstruction. Adding multiple attention modules here leads to a rise in computational costs without providing proportionate performance gains. In contrast, skip connections bridge low-level spatial details with high-level semantic cues. Integrating attention at the skip connections enhances focus on critical regions while limiting computational overhead, ultimately improving segmentation performance.

Based on this rationale, we conducted ablation studies incorporating attention mechanisms into the skip connections. To maintain a single varying factor, VGG16 + UNet was chosen as the base network (denoted as Base), and the performance of various attention modules integrated into the skip connections was evaluated. The models tested include Base + SE (squeeze-and-excitation), Base + SA (ShuffleAttention), Base + CBAM (convolutional block attention module), and Base + PSA (pyramid squeeze attention, our proposed module). All experiments were performed on the MPDataset, and the results summarized in Table IV indicate that integrating SE or CBAM modules into the skip connections leads to slightly inferior segmentation performance compared to other configurations. In contrast, when the PSA mechanism is incorporated into the skip connections, it leads to a noticeable improvement in the model's performance. When Base + PSA is compared to Base + SA on the MPDataset, it outperforms it in all evaluation metrics with MIoU, MPrecision, recall, and MPA increasing by 0.31%, 0.65%, 0.68%, and 1.26%, respectively. These improvements suggest that the PSA mechanism, which focuses on efficiently capturing multiscale spatial information, is more effective at enhancing the model's segmentation capabilities than other attention mechanisms.

C. Extension of the MPSUNet Model to Sentinel-2 Data

The MPSUNet model trained on the MPDataset was extended to a Sentinel-2 dataset published by Irakulis-Loitxate et al. [41], to further evaluate its generalization capabilities and adaptability for different satellite data sources. This Sentinel-2 dataset was manually annotated and consisted of 10 046 images with a spatial resolution of 200×200 pixels, generated from Sentinel-2 scenes centered on 23 known super-emitter locations in Turkmenistan during the period from 2017 to 2021. Each data point in the Sentinel-2 dataset includes derived methane enhancement data and the corresponding standard RGB data, and its structure is the same as our EMIT MPDataset.

Despite intersensor variability, cloud interference, and inherent noise in Sentinel-2 data, MPSUNet still maintained high detection performance, achieving an MIoU of 77.65% (78.20% for MPDataset), and the MPrecision and recall values reached 78.93% and 70.02% (80.78% and 71.11% for MPDataset), respectively, underscoring the model's accuracy and completeness in Sentinel-2 methane plume detection. The MPA

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TABLE IV
COMPARISON OF THE EFFECTS OF THE ATTENTION MODULE. BOLD INDICATES THE BEST RESULTS

Base	SE	SA	CBAM	PSA	MPDataset			
					MIoU(%)	MPrecision(%)	Recall(%)	MPA(%)
✓	✓				76.91	77.75	65.71	82.19
✓		✓			77.89	80.13	70.43	84.15
✓			✓		77.36	79.66	68.77	82.99
✓				✓	**78.20**	**80.78**	**71.11**	**85.41**

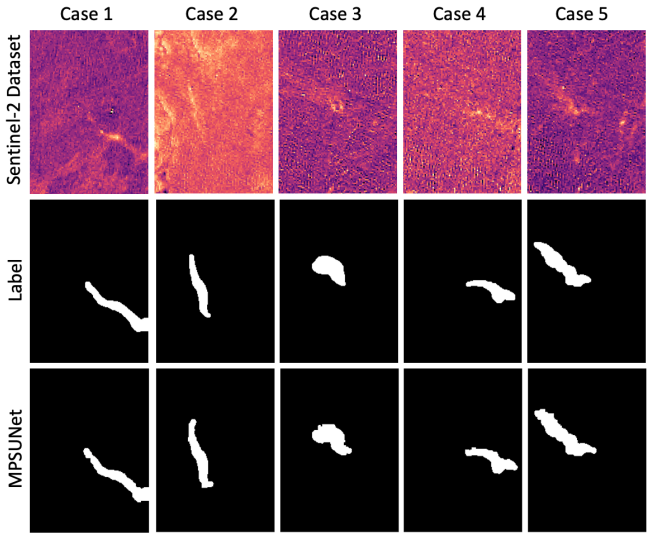

Fig. 11. MPSUNet prediction results on the Sentinel-2 dataset.

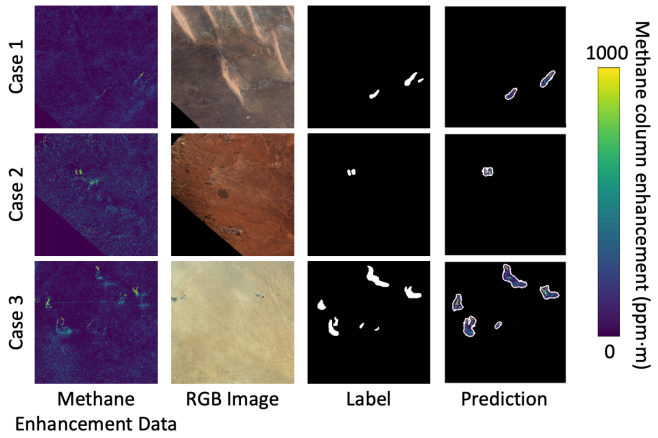

Fig. 12. Visual examples of failure cases in methane plume detection.

metric is 83.61% (85.41% for MPDataset), further reinforcing the model's overall performance. These results indicate that MPSUNet effectively generalizes to different satellite data sources and environmental conditions. Furthermore, Fig. 11 illustrates that the detected methane plume shapes and distributions are highly consistent with the ground-truth masks in the Sentinel-2 dataset, further reinforcing the model's reliability. The consistent performance across datasets indicates that MPSUNet is not overly sensitive to specific spectral characteristics, atmospheric conditions, or sensor-related noise. Instead, it effectively captures methane plume structures under diverse environmental settings, demonstrating its robustness to variations in terrain, cloud interference, and observational conditions.

D. Limitations and Future Directions

To better understand the limitations of MPSUNet, several failure cases are presented in Fig. 12. These examples highlight scenarios where small methane plumes were not effectively detected (Cases 1 and 3 in Fig. 12), and multiple closely located plumes were incorrectly merged into a single region (Cases 2 and 3). These issues primarily occur under complex backgrounds or where the methane signal is particularly weak. Such failure cases suggest that further improvements are still needed in enhancing the model's sensitivity to small-scale features and its ability to differentiate closely adjacent emission sources.

In consideration of these limitations, several future directions may help further improve model performance. Multimodal fusion [42] techniques, such as integrating wind data, land cover, or thermal infrared imagery, could provide complementary information to enhance plume localization, especially under low-signal or noisy conditions. In addition, spatiotemporal modeling frameworks [43] may enable the model to better track methane plumes across time series data, which is especially relevant for identifying intermittent emissions or correlating plumes with environmental factors. Furthermore, exploring latent feature modeling and graph-based architectures [44] may help improve the model's ability to distinguish between closely located or overlapping plumes by capturing high-order dependencies in spatial and contextual information. These directions represent promising avenues to enhance segmentation robustness and accuracy in real-world monitoring applications.

V. CONCLUSION

In this study, the application of deep neural networks for methane plume segmentation in remote sensing imagery was investigated. An improved segmentation model, MPSUNet, was proposed, which effectively integrates spatial and spectral information derived from the EMIT MPDataset to achieve robust plume detection under complex environmental conditions. MPSUNet incorporates the PSA module to enhance feature representation and employs a joint loss function combining Dice Loss and Focal Loss to address class imbalance while improving segmentation precision, particularly along complex plume boundaries. According to the prediction results, our main conclusions are summarized as follows.

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1) Based on the EMIT products, the high-quality MPDataset with 4172 samples was constructed by combining the methane enhancement data with RGB radiance bands. Experiments were conducted using the single-channel methane enhancement data (fourth channel of MPDataset) and the complete four-channel MPDataset, respectively. Our results indicate that the MPSUNet trained on the complete four-channel MPDataset outperforms the version trained on the single-channel methane enhancement dataset across all key segmentation metrics. The integration of RGB channels improved the segmentation quality, particularly in capturing small or intricate plume structures. The validation based on the emission source and wind field data further supports the reliability of MPSUNet in detecting authentic methane plumes, including smaller or undetected emissions overlooked by manual review.
2) Moreover, trained on the complete four-channel MPDataset, the performance of MPSUNet was further compared with the other seven deep learning models. Results indicated that our MPSUNet model outperformed all other models across all key metrics. Compared with other models, MPSUNet can capture fine-grained details and small-scale plume structures under complex scenarios.
3) In addition to its accuracy, MPSUNet maintains computational efficiency, with FLOPs and parameter counts comparable to or lower than other methods, such as Samba. This balance between accuracy and efficiency ensures that MPSUNet is not only effective but also practical for large-scale applications in satellite-based methane monitoring.
4) To further evaluate its generalization capabilities and adaptability of MPSUNet for different remote sensing data, the MPSUNet model trained on the MPDataset was extended to the Sentinel-2 data. This model still maintains high accuracy and robustness on Sentinel-2 data.

Our code can be accessible at https://github.com/MethaneTeam/mpsunet. Generally, the proposed MPSUNet model represents a significant advancement in the field of methane detection, offering a scalable, efficient, and accurate solution for addressing the urgent need for methane emission monitoring as emphasized by global climate agreements. Future work could focus on expanding the dataset to include more diverse environmental conditions beyond arid and semi-arid regions, particularly in humid and polar areas where diffuse methane emissions are prevalent. Additionally, incorporating auxiliary data such as wind speeds and land-cover information could further refine model performance and enhance its applicability to a broader range of methane emission scenarios.

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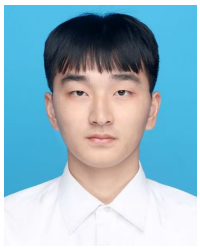

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